

DEVELOPMENT ENVIRONMENTS

Pull Requests on GitHub

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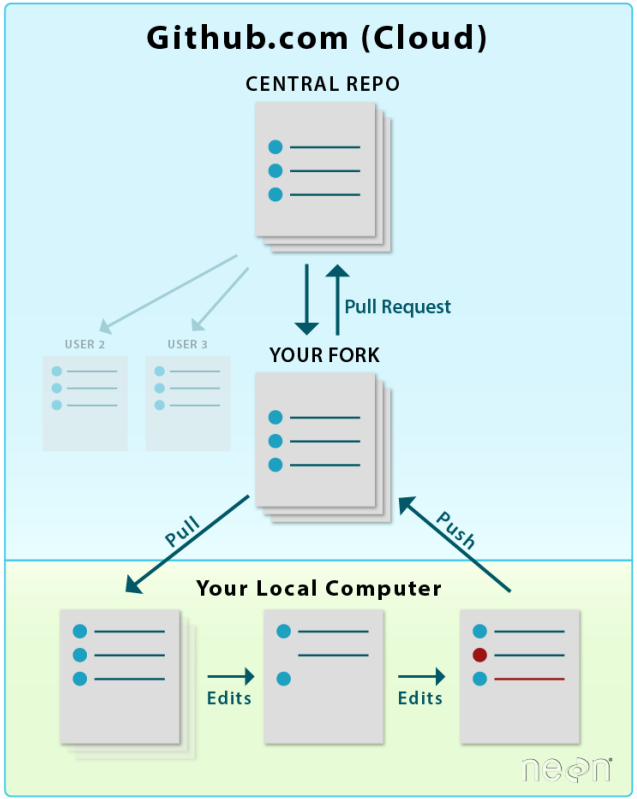
What is a Pull Request and when to use it?

A Pull Request is a request for another developer to incorporate a branch from your repository into theirs.

It is mainly used in the following scenarios:

- **When we do not have direct permissions to incorporate changes because we are not the owners of the repository.**
- **To create open source software and encourage collaborative work.**

The general flow consists of creating a branch, creating the Pull Request, and finally performing the branch merge.





PULL REQUEST

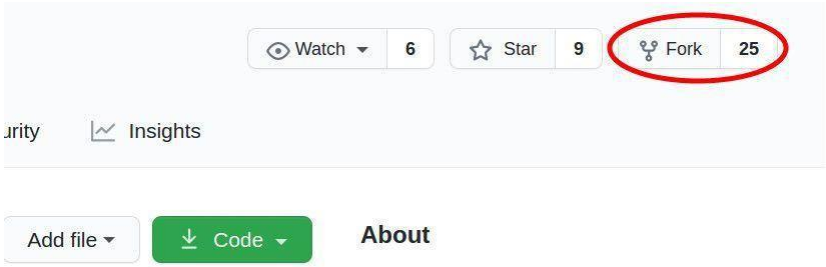
Steps as a collaborating programmer:

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Step 1: Fork the original repository

The first step to contribute is to fork the repository.

- Perform a fork by clicking the "Fork" button located in the top right corner of the project page on GitHub.
- This action will create an exact and complete copy of the original repository directly in your personal account.
- From this point forward, you have full control over this copy to experiment without affecting the original project.

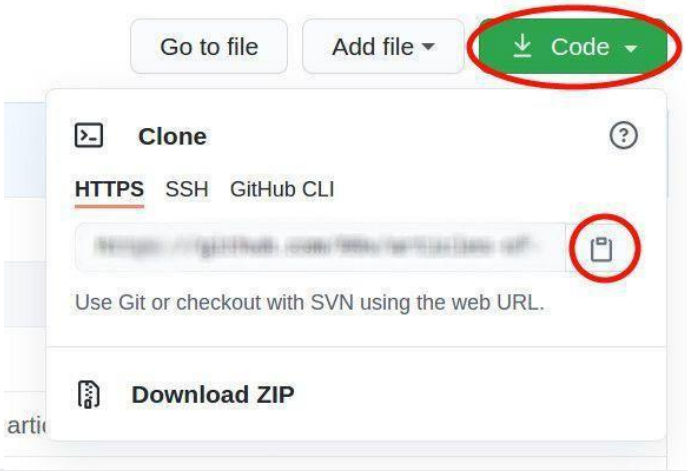


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Step 2: Clone the forked repository to your local machine

Once the repository is copied to your GitHub account, you need to clone it to your computer in order to work on it locally.

- **In your repository, click the green "Code" button and copy the link.**
- **Open your terminal and run the command clone to download the files:**




```
# Terminal
$ git clone [HTTPS ADDRESS]
```

3 Step 3: Create a new branch for your changes

It is good practice to create a new branch when working with repositories, whether in a small project or when contributing to a team.

The branch name should be short and clearly reflect the work being done.

Create a branch and navigate to it using the following commands:



Change Isolation

Keep the main branch clean while developing new features.

```
$ git branch [Branch Name]
$ git checkout [Branch Name]
```

4 Step 4:Make and commit changes using git add and git commit

Once you're in your new branch, make the essential changes to the project and save them to your files.

To prepare and confirm these changes to your local history, use the following commands:

- **Add the changes to the newly created branch:**

```
$ git add .
```

- **Confirm those changes with a descriptive message:**

```
$ git commit -m "Updating functionality"
```

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Step 5: Send the changes to GitHub with git push

After confirming the changes in your local environment, the next step is to upload them to your cloud repository.

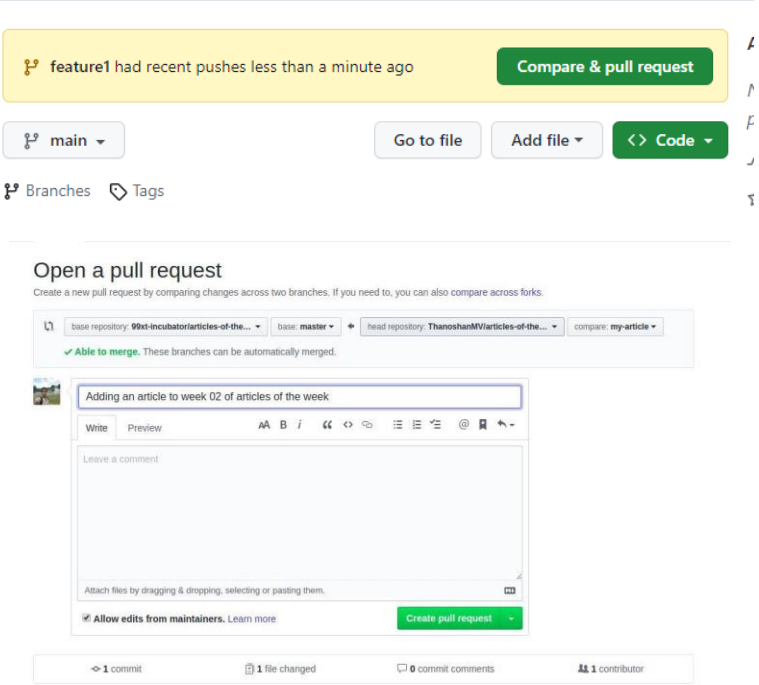
To push changes to GitHub, we need to identify the branch we're working on. Run the following command in your terminal:

```
$ git push origin [Branch Name]
```

6 Step 6: Open and detail the Pull Request on GitHub

Go to your repository on GitHub. When it detects your new branch, you'll see a button called "Compare & pull request".

- Click the button to start the application.
- Provide all the necessary and descriptive details of what you have done in the form.
- Send the pull request by clicking on"Create pull request".



> *Congratulations! You have submitted your first pull request. If it is accepted, you will receive a confirmation email.*



PULL REQUEST

Steps as repository owner/administrator:

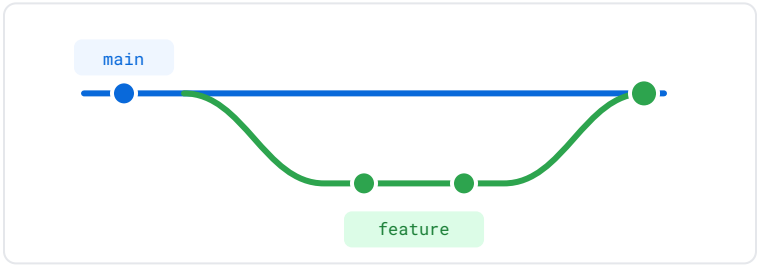
Code review and merge by the owner

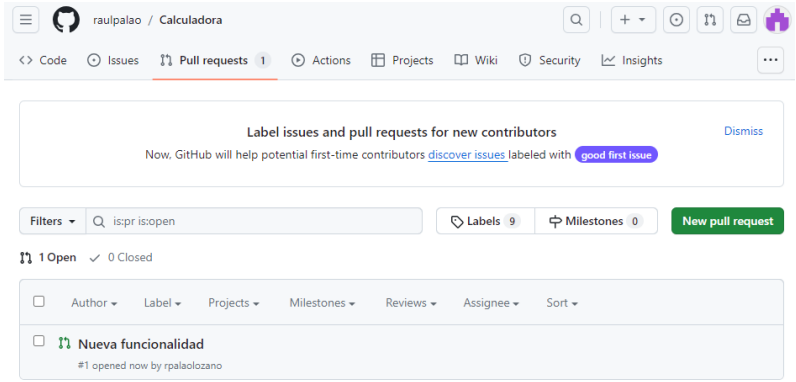
Once the Pull Request has been sent, the responsibility passes to the administrator of the original repository.

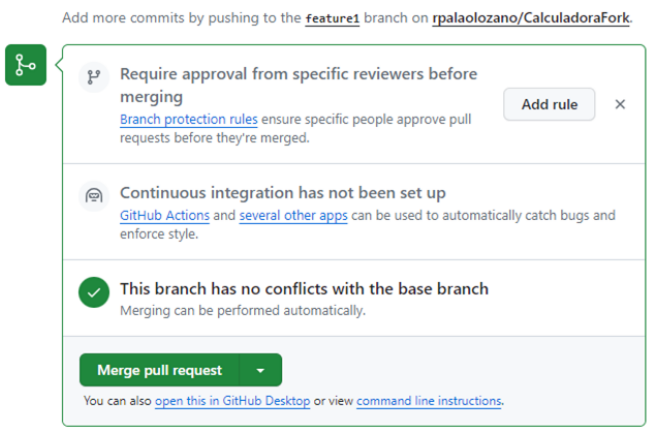
1. The project owner can review all submitted requests by navigating to the tab `"Pull requests"`.

2. If the code does not conflict with the base branch, it will be indicated that the `"Merging can be performed automatically"`.

3. The maintainer will then merge your changes into their branch `"main"`, thus integrating your contribution into the official project.









PULL REQUEST

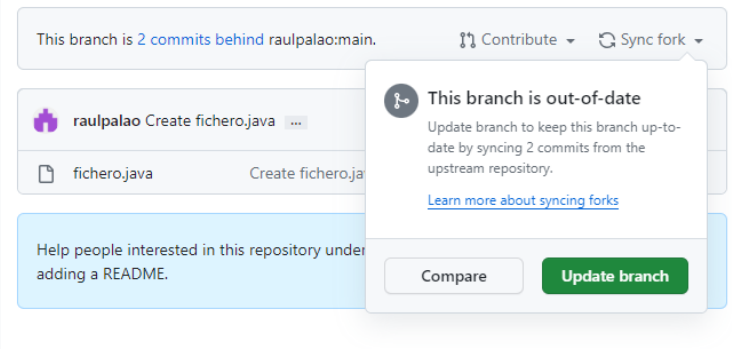
Another contributionas a collaborating programmer:

Synchronization of Forks and updating of the local repository

To continue collaborating, you must keep your repository updated with the original:

1. Update your fork on GitHub:

- Use the option "Sync fork“ and click on "Update branch“ to synchronize the commits from the original repository (upstream) with your main branch.



2. Update your local environment:

- Switch to the local master branch:

```
$ git checkout main
```

- Download the updated changes:

```
$ git pull origin main
```

Another new contribution

To make a new contribution, we repeat the entire process from a synchronized environment:

1 **Create a branch:**

```
$ git branch [Name] and $ git checkout [Name]
```

2 **Make and confirm changes:**

```
$ git add . and $ git commit -m "Message"
```

3 **Send to GitHub:**

```
$ git push origin [Name]
```

4 **Create Pull Request:**

Use the "Compare & pull request" button on GitHub.



All that's left is for the project owner to accept the changes and perform the merge.